RV College of Engineering®, Bengaluru
(Autonomous Institution affiliated to VTU, Belagavi)
Department of Industrial Engineering & Management

Manufacturing Process Lab
IM234AI

Laboratory Manual (Autonomous Scheme 2022)

RV College of Engineering®, Bengaluru
(Autonomous Institution affiliated to VTU, Belagavi)
Department of Industrial Engineering & Management

Laboratory Certificate

This is to certify that Mr. / Ms.…… ……
……has satisfactorily completed the course of Experiments in Practical …… prescribed by the Department during the academic year …….

Name of the Candidate: ……...……..

USN No.: ……..……..…… Semester: …….…….

Marks	
Maximum	**Obtained**
50	

Marks in words	

Signature of the Staff in-charge with date

Head of the Department

RV College of Engineering®, Bengaluru
(Autonomous Institution affiliated to VTU, Belagavi)
Department of Industrial Engineering & Management

MANUFACTURING PROCESS (IM234AI)
UNIT-VI (Laboratory Work)

SCHEME OF CONDUCT AND EVALUATION

CLASS: III SEMESTER
YEAR: 2025_26

CIE MARKS: (Max.) : 50
SEE MARKS: (Max) : 50
SEE: 03 Hrs

Expt. No.	TITLE	No. of Classes	Max. Marks
1.	Sketches & uses of tools used in Foundry & Machine shop	1	20
Part - I - Sand Testing / Foundry			
2.	Determination of Grain Fineness Number (GFN) of given clay free sand sample using Sieve shaker.	1	20
3.	Evaluation of the Effect of moisture content on shear strength of given green sand.	1	20
4.	Preparation of core without using a pattern- Core making.	1	20
5.	Preparation of Split (Two pieces) Pattern - Pulley.	1	20
Part – II – Machine shop			
6.	Plain Turning, Grooving, Taper turning, Knurling, External thread cutting	1	20
7.	Plain Turning, Grooving, Knurling, Chamfering & Forming (Concave & Convex Turning)	1	20
8.	Spur Gear cutting using milling machine	1	20
9.	Helical Gear Cutting using milling machine	1	20
10.	Demonstration of Surface Grinding machine Demonstration of Non Destructive Testing Equipment's (Ultrasonic & Magnetic Particle Inspection)	1	–
Part – III – Innovative Exercises			
11.	Project based learning	1	20
TEST		**1**	**50**
TOTAL			250

EVALUATION SCHEME:
CIE Marks = {Total Class marks (Max.200) + Test marks (Max.50)} Reduced to 50 Marks

***Reference for conceptual foundation**: P.N. Rao, Manufacturing Technology: Foundry Forming and Welding, TMH, 2nd Edition, 2003, ISBN: 0-07-463180-2.*

Particulars of the Experiments Performed

CONTENTS

Sl. No.	Date	Experiments	Page No.	Marks Obtained	Signature of Staff
Part - I - Sand Testing / Foundry					
1.		Sketches & uses of tools used in Foundry & Machine shop	**1 – 32**		
2.		Determination of Grain Fineness Number (GFN) of given clay free sand sample using Sieve shaker.	**33 – 37**		
3.		Evaluation of the Effect of moisture content on shear strength of given green sand.	**38 – 41**		
4.		Preparation of core without using a pattern-Core making.	**43 – 44**		
5.		Preparation of Split (Two pieces) Pattern - Pulley.	**45 – 46**		
Part – II – Machine shop					
6.		Plain Turning, Grooving, Taper turning, Knurling, External thread cutting	**47 – 48**		
7.		Plain Turning, Grooving, Knurling, Chamfering & Forming (Concave & Convex Turning)	**49 – 50**		
8.		Spur Gear Cutting using Milling Machine	**51 – 52**		
9.		Helical Gear Cutting using Milling Machine	**53 – 54**		
10.		Demonstration of Surface Grinding machine	**55 – 57**	--	
		Demonstration of Non Destructive Testing Equipment's (Ultrasonic & Magnetic Particle Inspection)	**58 – 59**	--	
Part – III – Innovative Exercises					
11.		Project based learning	**60 – 61**		
Test					
Total					

CIE Marks (Reduced to 50)

= {Total Class marks (Max.200) + Test marks (Max.50)}/ 50

Common Instructions

- ⇨ The Students are required to strictly follow the scheme of conduction of the experiments.
- ⇨ The Students are hereby advised to prepare for the Experiments well in advance and the timing of the lab slots are to be effectively utilized for the computational and analysis aspects of the experiments.
- ⇨ The Students are required to bring in their Blue Books duly complete in all respects, without which the students will not be allowed to do the experiments in the lab.
- ⇨ Submission of Lab Records complete in all respects is to be done in the next subsequent Lab Class.
- ⇨ Students must maintain strict academic discipline in the Laboratory.
- ⇨ Use of external hard disks (thumb drives) in the Lab without prior permission of the Lab-in-charge will be viewed very seriously.
- ⇨ CIE Marks will be awarded based on Conduction, Analysis, Result & Inference with necessary attendance requirement.
- ⇨ **Open ended** and **Demo** experiments are considered **only for CIE marks** and not for **SEE** consideration.

Safety Measures

- ⇨ Keep proper discipline in the shop and no horseplay.
- ⇨ Don’t play with the tools because they are sharp and you may get hurt.
- ⇨ Don’t stand in front of the person who is forging.
- ⇨ Leather shoes must be worn and hand gloves.
- ⇨ If any accidents happened do not secret it always inform your instructor or concerned staff.
- ⇨ Don’t run any machine unless you are sure how it operates.
- ⇨ Protect your eyes by wearing goggles wherever necessary.
- ⇨ Don’t wear any necktie, jewellery, rings and watches during the working in the shop.
- ⇨ Keep your working place free from oil and grease.

Vision

Imparting innovation and value based education in Industrial Engineering and Management for steering organizations to global standards with an emphasis on sustainable and inclusive development.

Mission

1. To impart scientific knowledge, engineering and managerial skills for driving organizations to global excellence.
2. To promote culture of training, consultancy, research and entrepreneurship interventions among the students & faculty.
3. To institute collaborative academic and research exchange programs with National and globally renowned Universities, industries and other organizations.
4. To establish and nurture Center of Excellence in the niche area of Industrial and Systems Engineering.

Program Educational Objectives (PEOs)

BE in Industrial Engineering and Management program, graduates will be able to:

i. Conceive, design, implement and operate integrated systems, focus on appropriate measures of performance at strategic, tactical and operational levels.
ii. Develop competency to adapt to changing roles for achieving organizational excellence.
iii. Design and develop sustainable technologies and solutions for betterment of society.
iv. Pursue entrepreneurial venture with a focus on creativity and innovation for developing newer products, processes and systems.

Program Outcomes (POs)

Engineering Graduates will be able to:

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and Industrial Engineering concepts to the solution of complex engineering problems.

2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

6. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

11. **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific outcomes

Engineering Graduates of IEM will be able to:

1. Design, develop, implement and improve integrated systems that include people, materials, information, equipment and energy.

2. Apply statistical and simulation tools, optimization and metaheuristics techniques for analysis of various systems leading to better decision making.

3. Demonstrate the engineering relationships between the management tasks of planning, organization, leadership, control, and the human element in various sectors of economy.

Rubrics for Sketches:

Parameters	Points considered for evaluation	Scale for evaluation				
		4	**3**	**2**	**1**	**0**
Conceptual Understanding	1. Ability to identify the tools used in foundry and forging process 2. Usages of tools in the foundry and forging process. 3. Knowledge of using tool with safety precautions.	• Able to identify all the tools used in foundry and forging process. • Explain usages of tools in the foundry and forging process. • Knows using tools with safety precautions.	• Able to identify most of the tools used in foundry and forging process. • Explain usages of tools in the foundry and forging process. • Knows using tools with safety precautions.	• Able to identify most of the tools used in foundry and forging process. • Cannot explain usages of tools in the foundry and forging process. • Knows using tools but do not follow safety precautions	• Able to identify very few of the tools used in foundry and forging process. • Cannot explain usages of tools in the process. • Knows using tools and but do not follow safety precautions.	• Unable to identify any of the tools used in foundry and forging process. • Cannot explain usages of tools in the foundry and forging process.
Sketches and Applications	1. Neat sketch with complete labeling. 2.Ability to apply concept for engineering problems.	**8**	**7-5**	**4-3**	**2-1**	**0**
		Sketch is complete in all aspects & able to apply concept to various engineering problems.	Sketch with few labeling are missing and lack of application to problems.	Sketch with more than two labeling are missing Improper justification to applications.	Poor sketch and no application	No sketch No application
Viva	1. List all the tools used in foundry and forging process 2. Explain how to use tools 3. Explains safety instruction to be followed while using tools.	**3**	**2**	**1**		**0**
		Explain all tools used and thorough knowledge of safety instructions in foundry and forging process	Explain most of the tools used and moderate knowledge of safety instructions in foundry and forging process	Explain few of the tools used and poor knowledge of safety instructions in foundry and forging process		Cannot explain the tools used and do not know safety instructions in foundry and forging process

Rubrics for Lab experiments: (20 Marks)

Parameters	Points considered for evaluation	Scale for evaluation				
		4	**3**	**2**	**1**	**0**
Conceptual Understanding	1. Conceptual understanding of the experiment 2. Clarity of purpose of conducting experiment 3. Has clear knowledge of conduction of experiment,	Thorough understanding of principles & Procedure of experiment	Moderately understand the principles & Procedure of experiment	Do not understand the principles but Explains Procedure of experiment	Do not understand the principles but Explains few steps of experiment	Cannot explain principles & Procedure of experiment
Conduction of the experiment	1. Understand the objective of experiment. 2. Conduct the experiment independently.	**5**	**4**	**3**	**2-1**	**0**
		Conduct the experiment correctly with all steps	Conduct the experiment without knowing purpose	Conduct the experiment with more than two steps missing	Conduct the experiment without following any steps	No experiment conducted
Analysis Results and Inference	1. Complete analysis of the given experiment 2. Understanding the nature of graph. 3. Articulation of answers	**3**		**2**	**1**	**0**
		Completed analysis, Interpretation complete in all aspects		Partially completed analysis, Interpretation complete in all aspects	Partially completed analysis, No Interpretation	No analysis, No Interpretation
Viva	1. Explains questions related to Principles, procedure and applications. 2. Clarity of communication of ideas and thoughts.	**3**	**2**	**1**	**0**	
		Answering all questions correctly	Does not answer 1 or 2 questions	Does not answer few questions	Does not answer any question	

Rubrics for Lab experiments: (20 Marks)

Parameters	Points considered for evaluation	Scale for evaluation				
		4	**3**	**2**	**1**	**0**
Conceptual Understanding	1. Conceptual understanding of the experiment 2. Clarity of purpose of conducting experiment 3. Has clear knowledge of conduction of experiment,	Thorough understanding of principles & Procedure of mold making	Moderately understand the principles & Procedure of Mold making	Do not understand the principles but Explains Procedure of mold making	Do not understand the principles but Explains few steps of molding	Cannot explain principles & Procedure of molding
Conduction/ Making mold	1. Understand the objective of mold making. 2. Conduct /make mold independently.	**4**	**3**	**2**	**1**	**0**
		Conduct the make mold correctly with all steps	Molding without knowing the purpose	Molding with more than two steps missing	Molding without following any steps	Mold not made / Unable to make mold
Dimensional Accuracy	Labeling with accurate values	**4**	**3**	**2**	**1**	**0**
		Highly accurate values	Moderately accurate	Good accurate values	Poor accuracy	Least accuracy
Finish	Surface finish of mold cavity /core	**4**	**3**	**2**	**1**	**0**
		Very good surface	Good finish surface	Coarse finish surface	Rough surface	Very rough surface
Viva	1. Explains questions related to Principles, procedure and applications. 2. Clarity of communication of ideas and thoughts.	**4**	**3**	**2-1**	**0**	
		Answering all questions correctly	Does not answer 1 or 2 questions	Does not answer few questions	Does not answer any question	

CONCEPTUAL FOUNDATION

Moulding Sands

The common sources of collecting foundry sands are rivers, lakes, sea and deserts. All the foundry sands can be mainly grouped as:

1. **Natural sand** – which contains sufficient amount of binding clay and, therefore, no more binder is required to be added to them.

2. **Silica sands** – which do not posses the clay content and need addition of a suitable binder to make them usable for foundry work. They are also sometimes called sharp sands. They, when mixed with some other constituents like binders and additives, are also known as synthetic sands.

All foundry sands; owe their colour to the impurities present in them. The common colours of sands used in foundry work are white, yellow, brown, grey and red. Silica is usually the main constituent of all the sands; irrespective of their colour and the source of availability.

Characteristics of Foundry sands

1. **Refractoriness:** It is that property of the moulding sand which enables it to withstand high temperature of the molten metal without fusing, thus facilitating a clean casting. The extent to which this property is needed depends upon the metal which is to be cast. For example, steel is poured at a higher temperature than cast iron. Similarly, cast iron is poured at a higher temperature than brass. Therefore, the mould for steel casting should have higher degree of refractoriness than that used for iron castings and the latter should have more refractoriness than the one used for brass casting. If sand lacks in this property it will fuse on, coming in contact with the molten metal, from slag on the surface of the casting and will, thus, spoil the latter.
   The moulding sand owes this property largely to the presence of quartz content or SiO_2 and the shape and size of its grains. Rough & larger grains will leads to higher refractoriness. On the other hand iron oxides, calcium, sodium and potassium react chemically, during pouring of molten metal, to form compounds which have very low melting points. Thus, their presence in the sand reduces its refractoriness in those sands which are used in iron and steel foundries.
2. **Permeability:** It is also termed as porosity. It is that property of the sand which allows the gases and steam to escape through the sand mould. When the hot molten metal is poured in the mould a very large volume of gases and steam is formed due to heating of moisture, coal dust, oil and similar other materials present in the sand. If these gases are not allowed to go out they will either make the casting unsound or blast the mould. Therefore, this is a very important property required in the moulding sand. It largely depends upon the sand grain size and shape and the proportion of moisture and clay present in the sand. Rounded grains of uniform size lead to higher permeability. This property is also affected by ramming of sand. A soft ramming will increase the

permeability and hard ramming will reduce it. In practice, it is further increased by applying vent wires in the prepared mould.

3. **Flow-ability or Plasticity:** It is that property of the sand due to which it flows during ramming to all portions of the moulding flask, packs properly around the pattern to acquire the desired shape and distributes the ramming pressure evenly to all parts of the mould. It increases with the addition of clay and water content and reduction of green strength and grain size.
4. **Adhesiveness:** It is that property of the sand due to which it is capable of adhering to the surfaces of other materials. It is entirely due to this property that the heavy sand mass is successfully held in moulding flask and manipulated as desired without any danger of its falling down. Similarly, on account of the same property, the gaggers are able to hold bulky sand projections of the mould.
5. **Cohesiveness**: It is that property of the sand due to which its rammed particles bind together firmly so that the pattern is withdrawn from the mould without damaging the mould surfaces and edges. Also, due to this property the mould faces get sufficient strength to withstand the pressure of the following molten metal and do not get washed under this pressure. This property of the sand in green state is known as green bond or green strength and is largely affected by the grain size, clay and moisture content.

   When the molten metal is poured into the mould or the latter is baked in an oven, it gets dried due to the evaporation of the moisture. The cohesiveness in sand grains is still required to give sufficient strength to the mould faces to retain their shape and resists the flow of metal into the sand mass. This property in dry state of the sand mould is known as dry bond or dry strength and depends upon grain size, binder and moisture content. The dry strength should, however, not allowed exceeding beyond requirements otherwise it will reduce collapsibility and restrict contraction of solidifying metal, resulting in cracks and unsound castings. It is particularly important in case of steel castings and iron castings having thin sections.
6. **Collapsibility:** It is the property due to which the sand mould automatically collapses after solidification of the casting to allow a free contraction of the metal. In absence of this property the contraction of the metal will be hindered by the mould and this will result in tears and cracks in the casting.
7. **Other requirements:** In additional to the above main characteristics, the moulding sand should also possess the following properties:
   i. It should be cheap and easily available.
   ii. Its coefficient of expansion should be sufficiently low.
   iii. It should be reusable to effect economy.
   iv. It should not react chemically with the molten metal.
   v. It should not stick to the surface of the castings.

Terminology of Foundry Sands

Different terms are commonly used to denote foundry sands in different conditions and having different uses. These terms are the following:

1. **Green sand.** It is also known as tempered sand. It denotes well prepared foundry sand which contains just enough moisture to give it sufficient bond. Moulds in this sand are known as green sand moulds and do not require any baking before pouring the molten metal into them.

2. **Dry sand.** This term indicates that moulding sand which was originally having excess moisture content but the same has been evaporated from it by drying its mould in a suitable oven.

3. **Facing sand**. It is also known as 'fat' sand. These terms are used for that sand which forms the face of the mould, i.e., rammed around the pattern surface. It is nothing but the fresh prepared and well tempered foundry sand. Initial coating around the pattern surface is given by this sand and the remainder of the flask is filled with floor sand to effect economy.

4. **Parting sand.** This term denotes that sand which is sprinkled on the pattern and the parting surfaces of the mould so that the sand mass of one flask does not stick to that of the other or to the pattern.

5. **Floor, black or baking sand**. These are interchangeable terms and all denote the used sand, which is left on the floor after the castings have been removed from the mould. Before reusing, it is riddled to remove foreign material like nails and fins etc. and then used for filling the bulk of the moulding flask after the facing sand has ken rammed around the pattern.

6. In modem mechanized foundries, however, no facing sand is prepared separately, but the entire floor sand is riddled, added with binders and proper additives and properly tempered for being used again. Such sand is called 'unit sand'.

7. **Core sand.** The sand, which carries high silica content and is used for making cores, is known as core sand.

8. **Oil sand.** Silica sand using oil binders is known as oil sand.

9. **Molasses sand.** This term denotes the sand, which carries molasses as binder. It is very useful for making moulds of small castings having intricate shapes and thin sections. Also it is used as core sand.

Sand Grains

The shape and size of the sand grains has a remarkable effect on the physical properties of the foundry sand. The sand grains may have smooth or rough surfaces. Out of these the first type. i.e., smooth, is preferred for moulding for the reason that such a surface renders higher permeability, sinter point and plasticity to the sand mass, but the percentage of binder required is also equally high.

Similarly the sand grains may have different shapes. The commonly formed shapes are rounded, sub-angular, angular and compound. The rounded grains do not bind together too well when rammed and, hence render the sand mould highly permeable but the strength of the mould is also reduced.

Sub-angular grains give a relatively stronger bond than above but the permeability is reduced. Angular or sharp grains produce a much stronger bond and a low permeability when rammed. Thus they enable a mould of greater strength. Sand grains which 'are cemented together such that they do Dot separate when screened are called compound. They may consist of one, two or a combination of all the above three shapes. They are not much preferred.

Like the shape the size of sand grains also affects the mould structure and its characteristics. Large, regular and uniform grains increase permeability. Smaller grains increase smoothness on mould surfaces.

Selection of Foundry Sand

Light or fine-grained sand is used for light castings, which carry intricate and finely carved details. The sand should carry a strong bond. Medium grained sand is preferred in bench work and light floor work. Heavy or close sand is widely used for large iron and steel castings. This sand should carry a high silica content, strong bond and low proportion of lime.

Main Constituents of Moulding Sand

The principle constituents of moulding sand are:

1. Silica sand,
2. Binder,
3. Additives,
4. Water.

1. **Silica sand** – which do not posses the clay content and need addition of a suitable binder to make them usable for foundry work. They are also sometimes called sharp sands. They, when mixed with some other constituents like binders and additives, are also known as synthetic sands

2. **Binders:** The purpose of adding a binder to the moulding sand is to import is sufficient strength and cohesiveness so as to enable it to retain its shape after the mould has been rammed and the pattern withdrawn. However, it produces an adverse effect on the permeability of the sand mould.

   The common binders used in foundry can be grouped as:

   i. Organic binders
   ii. Inorganic binders.

i. Organic binders find their specific use in core making. The common binders coming in this category are:

a) Dextrin
b) Molasses
c) Linseed oil
d) Cereal binders
e) Pitch -upto 2% max
f) Resins, like phenol and urea formaldehydes

ii. In the inorganic group the common binders are clay, sodium silicate and Portland cement, Out of all these, the clay binders are commonly used. The following types of clays are commonly used:

a) Bentonite
b) Kaolonite
c) Limonite
d) Ball clay
e) Fire clay
f) Fuller's earth.

Bentonite is most widely used among the above six varieties.

3. Additives

Additives are those materials which are added to the moulding sand to improve upon some of its existing properties or to impart certain new properties to it. The commonly used additives are

i. **Coal dust.** It is mainly used in the sand used for grey iron and malleable iron castings. Its main purpose is to react chemically with the oxygen present in the sand pores and, thus, produce a reducing atmosphere at mould-metal interface and prevent oxidation of the metal. For this reason its major portion is added in the facing sand. It however reduces the cohesiveness and strength of the sand.

ii. **Sea coal** It is a finely ground soft coal and is vastly used in sands used for grey and malleable iron castings. It restricts the mould wall movement and improves surface finish. It reduces permeability and hot strength of the mould and requires a higher percentage of water in the sand. Its proportion varies from 2 to 8 percent.

iii. **Cereals or corn flour.** It promotes mould wall movement by being volatized by heat, reduces expansion defects, improves strength, toughness and collapsibility of the sand and decreases permeability and flowability, its proportion in the sand varies from 0.25 to 2.0 percent.

iv. **Silica flour.** It increases hot strength, decreases metal penetration into the mould, reduces expansion defects and improves surface finish. It maybe added upto 35 percent.

v. **Wood flour.** It promotes mould wall movement, reduce-5 expansion defects; increases collapsibility improves surface finish and thermal stability of mould. It may be added from 0.5 to 2.0 percent or even more.

vi. **Pitch.** It improves hot strength and surface finish on ferrous castings. It can be advantageously added up to 2.0 percent if a higher proportion is added it will reduce green strength.

vii. **Dextrin and molasses.** Their addition increases the dry-strength of the sand. In other respects they behave more or less like corn flour.

viii. **Fuel oil.** Its, addition is sometimes done in order to reduce the requirement of free water in the sand.

4. Water

The clay content added to the foundry sand will not give the required strength and bond until a suitable quantity of water is added to it. This quantity of water varies from 2 to 8 percent according to different requirements.

The water content present in the sand mass is partly in mixed form, called pore water, and partly in the Free State, known as free water. When water is added to the clay it starts filling into the pores of the clay, where it forms a sort of micro- film. This water content is held rigidly by the clay and it is mainly responsible for enabling the clay to impart the desired strength to the sand. The quantity of bond and the capacity of clay for providing the bond will mainly depend upon the thickness of the water film that it can hold.

When more water is added to the clay mixture, than the amount required as pore water, it remains as a fluid and is held between the clay particles separating them. It has been found that this excess amount of free water behaves as a lubricant and, thus, improves the mouldability and plasticity of the moulding sand. It, however reduces the strength of the mixture and, thus, weakens the mould.

Experiment

01

Sketches of tools - Foundry and their uses:

Staff Signature

Sketches of tools used in Machine Shop **Sketch:**

Engine Lathe

Description: Main parts of Engine lathe

Sketch: Column & Knee Type Milling Machine

Description: Main parts of Column & Knee Type Milling Machine.

Sketches of lathe accessories:

Three Jaw Chuck	**Four Jaw Chuck**
Revolving Centre	**Drill Chuck with Key**
Catch Plate	**Carrier**
Steady Rest	**Follower Rest**

Description of Lathe Accessories:

Sketches of Cutting Tools:

V Tool	**Facing Tool**
Parting off Tool	**Knurling Tool**
Taper Shank Twist drill	**Involute Gear Cutter**

Description of Cutting Tools:

Sketches of Measuring Tools:

Outside Caliper	**Inside Caliper**
Steel Rule	**Outside Micrometer**
Vernier Caliper	

Description of Measuring Tools:

Sketches of Gauges:

Screw Pitch Gauge	**Standard Wire Gauge**
Cetre Gauge	**Feeder gauge**
Plug Gauge	**Ring Gauge**

Description of Gauges:

Scheme of Evaluation:

Particulars	Marks	Marks obtained	Staff Signature
Conceptual Understanding of all tools	**10**		
Sketch	**10**		
Total Marks	**20**		

Experiment

02

Grain Fineness Test (Sieve Analysis)

Aim: To determine the Grain fineness number of the given clay free sand sample

Apparatus: Sieve shaker with standard sieves attached with electric motor, wire brush, Stop watch, physical balance and pans. The sieves are of IS: 460-1962 specifications.

Theory: The grain fineness number of the sand is approximately 2.54 times the number of meshes per centimeter of that sieve which would allow passing sample if its grains were of a uniform size, i.e. the average of the sizes of the grain in the sample. It is approximately proportional to the surface area per unit weight of sand exclusive of clay. The grain fineness number test determines the grain size, grain distribution and grain fineness.

Sand grains could be fine, medium or coarse as regards size and could be rounded, semi angular or compounded as regards shape.

Granular particles of various sizes and shapes provide variable interstices (space between grains) and hence are directly responsible for permeability and compactness of the sand. Granular particles have higher strength but lower permeability; where as round grains have high permeability and lower strength.

Fine sand is desirable for intricate and small castings. As fine grains lie close, permeability is poor. Medium sand is used in bench work and light floor work. Coarse particles are used for large castings to permit gases to escape rounded grains have list contact with one another and lack in strength sub angular grains are comparatively less permeable than round grains . Angular grains have defined edges and thereby give more strength and less permeability

This number is a convenient means of describing the relative fineness of sand, most foundry sands being from about 40 to 220 in average fineness.

The sand properties however depend on the size distribution as well as on average size. The fineness test makes it possible to evaluate both the factors.

Procedure :(According to IS 1918-1966)

1. Sieve 50 grams of representative sand sample through 11 sieves starting with coarse sieve.
2. Using sifting time of not less than 15 minutes.
3. Weigh out the sand retained in each sieve and the pan individually and calculate percent retained.
4. Plot percentage retained vs. sieve size individually and cumulative.

Precautions:

i. Enough care should be taken in weighing the sand.
ii. All the sieves are to be arranged in proper sequence.

Experimental Setup:

The apparatus required for determining Grain fineness number consists of a number of standard sieves mounted one above the other, on a power driven shaker as shown in the figure. The sieves have different mesh sizes ranging from 3.35mm to 53 microns



SIEVE SHAKER

RV College of Engineering®, Bengaluru
(Autonomous Institution affiliated to VTU, Belagavi)
Department of Industrial Engineering & Management

OBSERVATION / DATA SHEET

Date: _____ Expt., No: _____

Title: Determination of Grain Fineness Number (GFN) using Sieve shaker.

Observations:

Weight of the sample (Sand) = 50 grams

Sifting time = 15 minutes

Sl.No	I.S. Sieve size	Multiplier (X)	Weight Retained In Grams	% Weight Retained (Y)	Cumulative % Weight Retained	X * Y
1	PAN	300				
2	53 µ	200				
3	75 µ	140				
4	106 µ	100				
5	150 µ	70				
6	212 µ	50				
7	300 µ	40				
8	425 µ	30				
9	600 µ	20				
10	850 µ	10				
11	1.70 mm	5				
12	3.35 mm	3				
					∑ X*Y =	

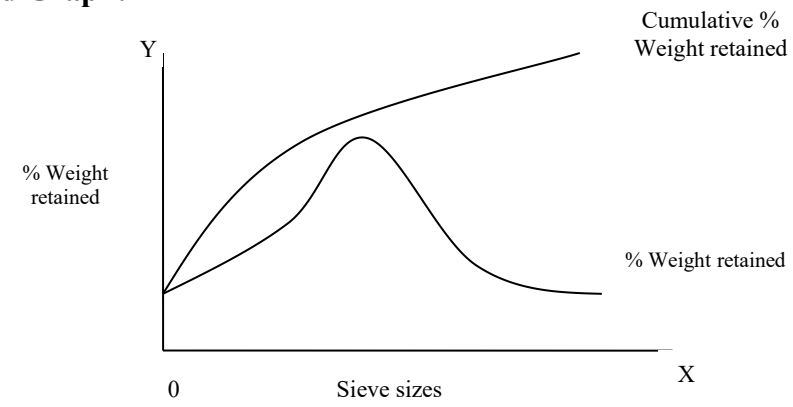
Formulae used:

1. % of sand retained in each sieve = $\frac{Weight\ of\ retained\ sand}{50} \times 100$

2. AFS Grain fineness number = $\frac{\sum(\%\ of\ weight\ retained\ \times multipler)}{Total\ \%\ weight\ retained}$

GFN =

Expected Graph:



Result and Discussion:

Scheme of Evaluation:

Particulars	Marks	Marks obtained	Staff Signature
Conceptual Understanding	**04**		
Conduction of experiment, Tabulation and calculation	**06**		
Analysis & Inference	**05**		
Viva	**05**		
Total Marks	**20**		

Insert Graph Sheet here

Experiment

03

Evaluation of the Effect of Moisture Content on Shear Strength for given Green Sand

Aim: To evaluate the effect of moisture content on green shear strength of a given sample of foundry sand.

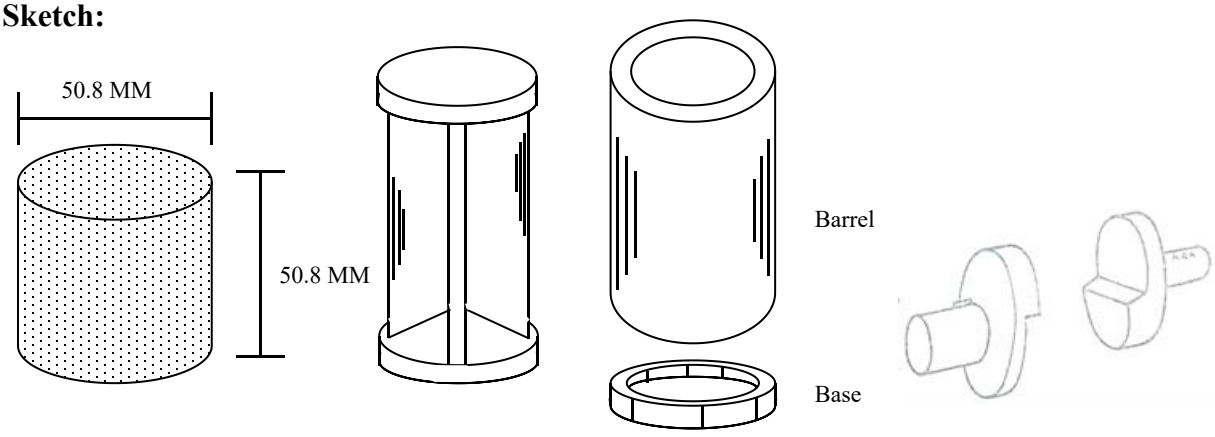
Apparatus / Equipment Used: Universal sand testing machine, shear shackles (heads), sand rammer weighing balance, water measuring jar.

Theory: The green shear strength of sand is the maximum shear stress which a tempered sand mixture is capable of developing, when prepared, rammed and tested according to standard procedure. Insufficient shear strength may lead to the collapse of the sand mould during handling the mould may be damaged during the flow of certain molten metal in the metal cavity. The moulding sand should have sufficient strength to permit the mould to be formed to desired shape and retain shape and size even after hot mould metal is poured in to the cavity.

Procedure: (According to IS: 1918 – 1966)

1. Ram a standard test specimen and strip from the specimen container.
2. Place the specimen between the shear heads of the machine in such a way that the load is applied along a line through its axis.
3. Apply a uniform load to the diametrically opposite halves of the two plain surfaces of the specimen at the rate of 17±4 gf /cm^2 per minute.
4. Record the shear strength in gf /mm^2.
5. Test three specimens individually. The shear strength shall be the average of three tests.
6. If the test results of one of the specimens varied more than 10 percent, from the average of the three, this result shall be discarded and another specimen tested.

Sketch:



Standard Specimen Stripper Mould Case Shackles

RV College of Engineering®, Bengaluru
(Autonomous Institution affiliated to VTU, Belagavi)
Department of Industrial Engineering & Management

OBSERVATION / DATA SHEET

Date: _____ Expt., No: _____

Title: **Evaluation of the Effect of Moisture Content on Shear Strength**

Observation:-

1. Varying the percentage of Clay:

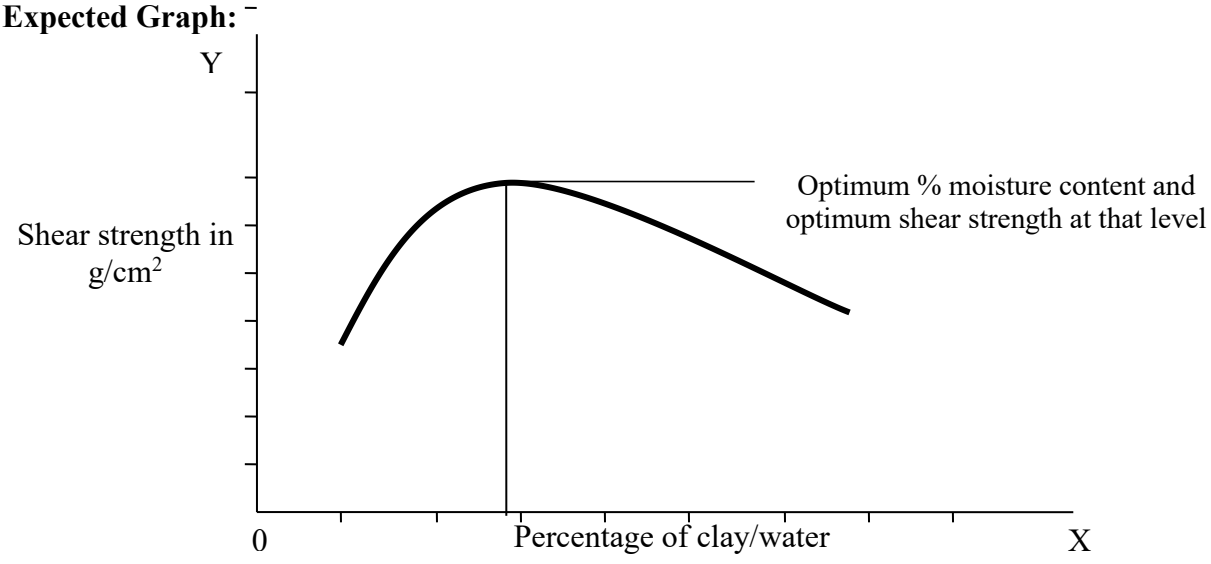
Sl. No.	Percentage of sand	Percentage of clay	Percentage of water	Shear strength gm/cm²
1.	91%(182gms)	4%(8gms)	5%(10gms)	
2.	90%(180gms)	5%(10gms)	5%(10gms)	
3.	89%(178gms)	6%(12gms)	5%(10gms)	

2. Varying the percentage of Water:

Sl. No.	Percentage of sand	Percentage of clay	Percentage of water	Shear strength gm/cm²
1.	90%(180gms)	6%(12gms)	4%(8gms)	
2.	89%(178gms)	6%(12gms)	5%(10gms)	
3.	88%(176gms)	6%(12gms)	6%(12gms)	

Staff Signature

Expected Graph:



Inference:

The Graphs above reveal:

1. With the increase in the percentage of water the shear strength of the specimen _____.

2. With the increase in the percentage of clay the shear strength of the specimen _____.

Scheme of Evaluation:

Particulars	Marks	Marks obtained	Staff Signature
Conceptual Understanding	**04**		
Conduction of experiment, Tabulation and calculation	**06**		
Analysis & Inference	**05**		
Viva	**05**		
Total Marks	**20**		

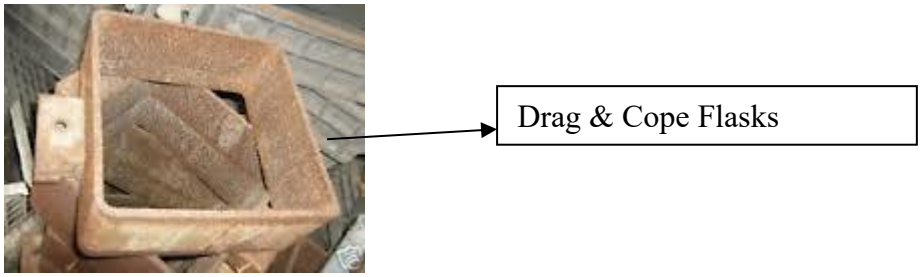
Insert Graph Sheet here

Operation Sheet for Foundry

Sl. No.	Description of Operation	Tools Used
1.	First the moulding box (drag) is taken. Then it's cleaned and then given a clay wash coating. Then the foundry sand is put into the drag box and filled to a height of about 2 inches above box.	Drag box, shovel
2.	Then with the peg rammer, ramming is done first along the corners and then on to the centers	Peg rammer
3.	After ramming with peg rammer, again foundry sand is filled up to 2 inches above the box then remaining is again carried out using round rammer to make the sand set firmly.	Round rammer
4.	After ramming with round rammer, leveling of the surface is done using levelers.	Levelers
5.	The leveled surface is smoothened using travels vent holes made.	Travels, Vent wires

Tools Used:





Experiment

04

Preparation of core without using a Pattern – Sand Moulding

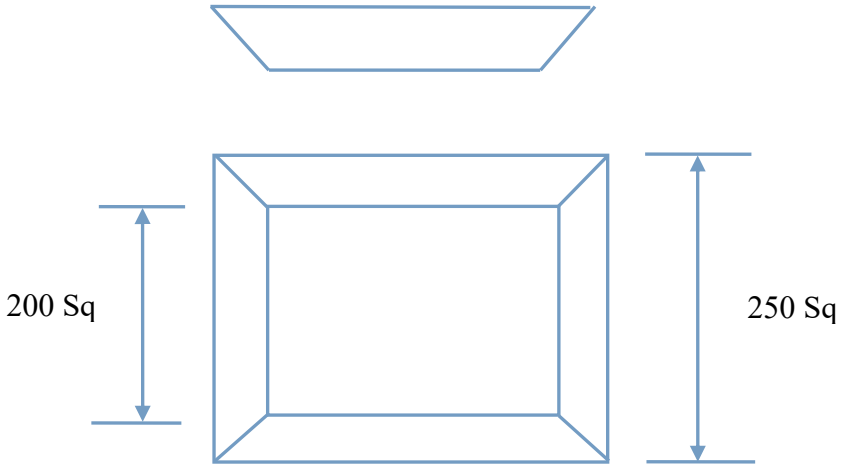
Aim: To Prepare sand mould without using a pattern (Hand Cutting).

Apparatus

- Moulding flasks, sand rammers, trowel, leveler, sand lifter,
- Vent rod and moulding sand mixture.

Procedure

1. The drag box is first smeared with Bentonite on its inner sides and then filled with moulding sand. It is first rammed with peg rammer twice. Then two inches height sand is placed over the drag and then it is subjected to ramming by a round rammer. Then the extra sand is removed using a leveler subjecting the top surface of the mould to a finish.
2. The cheek (cope) box is then placed over the drag. The surface of the drag is picked. The cheek is filled with the sand and the ramming process is repeated and a finished surface is got.
3. Sand is sprinkled on the top of the finished mould. The diagonals are drawn to get the center of the mould and the required shape [square] is drawn. The shapes are drawn using a vent rod.
4. The cheek (cope) box is removed carefully. Removing the sand using lifters and trowel gets the required shape. Care is taken to check the dimensions and using trowels finishes the surfaces. Thus the required mould is ready.



Sequence of Operations:

Model with Actual Dimension:

Scheme of Evaluation:

Particulars	Marks	Marks obtained	Staff Signature
Dimensional Accuracy	**08**		
Finish	**08**		
Viva	**04**		
Total Marks	**20**		

Experiment

05

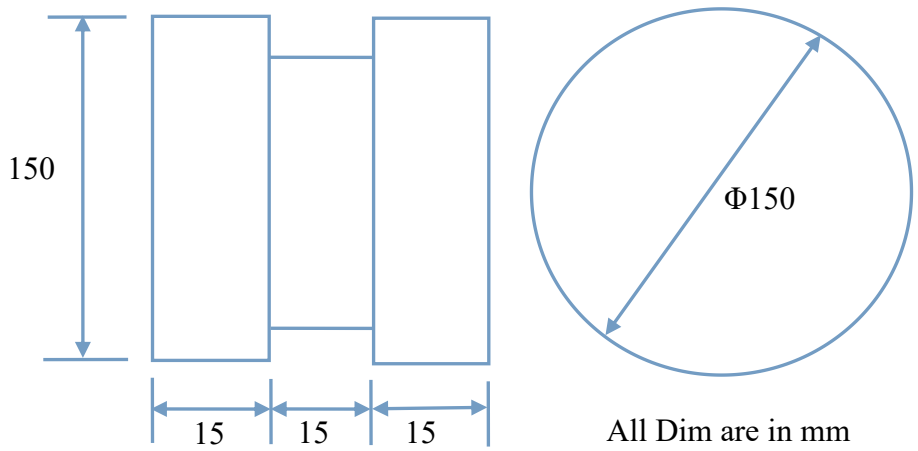
Preparation of Split (Two pieces) Pattern - Pulley

Aim: Preparation of Split (Two pieces) Pattern - Pulley.

Apparatus: Patterns, Cope and Drag, Rammers, Trowels, Levelers etc.

Procedure

1. Patterns are substances used to form cavities, so that moulds formed can be used for forming castings.
2. The mould making consists of two parts one in the cope and the other in the drag.
3. The part of the pattern is first placed in a drag sand is filled into the drag. The sand is rammed and leveled.
4. The drag is reversed. The runner is placed at two inches away from the core and not in the line of the core. Upon this another drag is placed and filled with sand with the other part of the pattern aligned with the first one.
5. Raiser is placed over the rim of the pattern. The sand is rammed and leveled. The surface is smoothened.
6. The drag is later taken of from the position. The runner and the raiser are removed.
7. The next procedure will be to remove the pattern. Since the sand will be tightly held to the pattern removing it is tough so the region around the pattern is wetted. A metal rod is fixed to the pattern.
8. The pattern is removed carefully by rapping it carefully. After both the patterns are removed gating system is provided. The prepared mould is allowed to bake and finally used for castings.



Sequence of Operations:

Model with Actual Dimension:

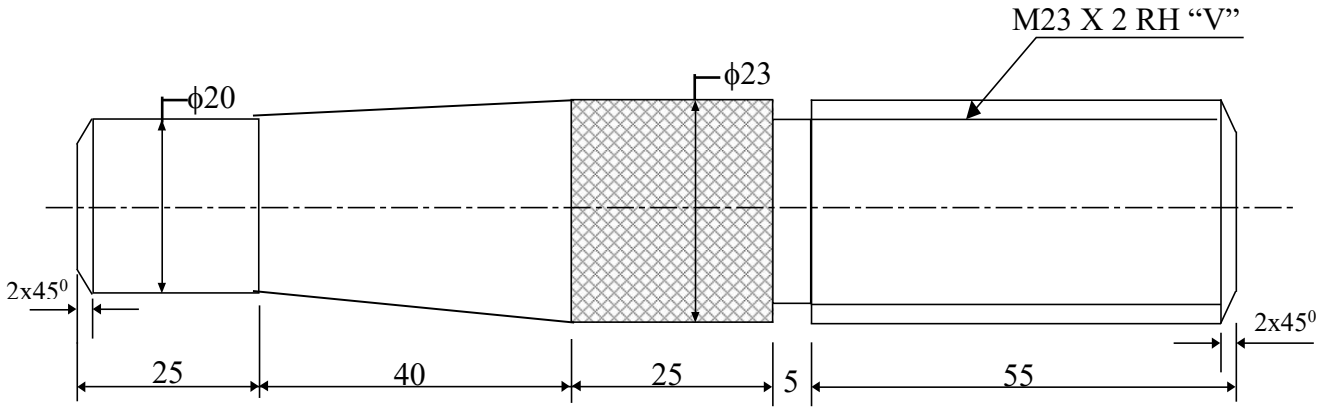
Scheme of Evaluation:

Particulars	Marks	Marks obtained	Staff Signature
Dimensional Accuracy	**08**		
Finish	**08**		
Viva	**04**		
Total Marks	**20**		

Experiment

06

Plain Turning, Grooving, Taper turning, Knurling, External Thread Cutting



All dimensions are in mm

Model with Actual Dimensions:

Operation Plan:

Sl No	Sequence of Operations	Tools Used
Measuring Instruments Used		

Given Dimension									
Measured Dimension									
Deviation									

Inference:

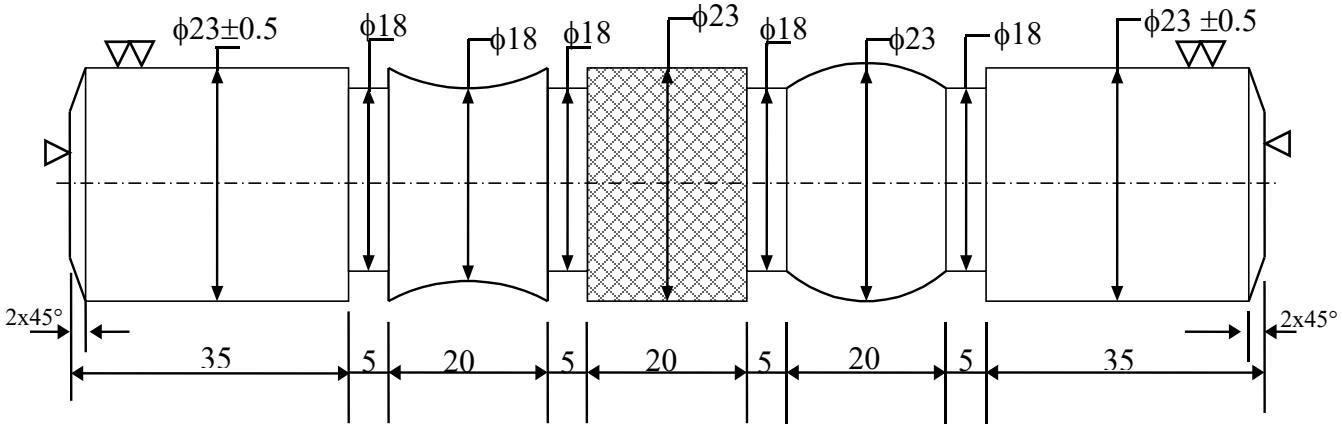
Scheme of Evaluation:

Particulars	Marks	Marks obtained	Staff Signature
Dimensional Accuracy	**05**		
Surface finish	**05**		
Report writing	**05**		
Viva	**05**		
Total Marks	**20**		

Experiment

07

Plain Turning, Grooving, Knurling, Chamfering & Forming (Concave & Convex Turning)



All dimensions are in mm

Model with Actual Dimensions:

Operation Plan:

Sl No	Sequence of Operations	Tools Used
Measuring Instruments Used		

Given Dimension									
Measured Dimension									
Deviation									

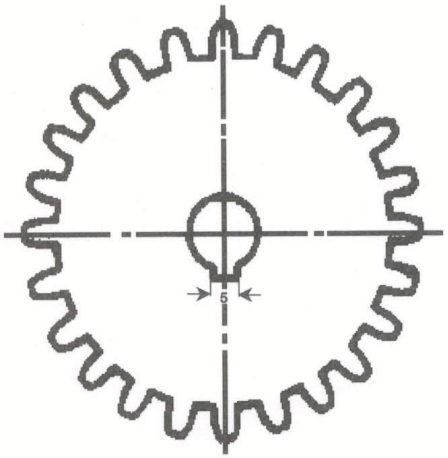
Inference:

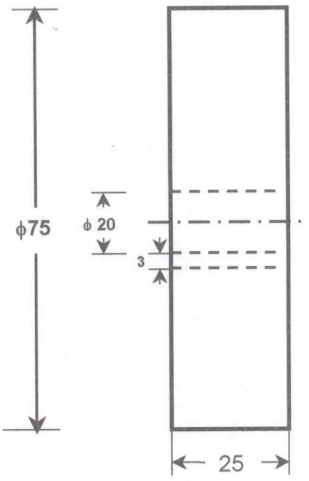
Scheme of Evaluation:

Particulars	Marks	Marks obtained	Staff Signature
Dimensional Accuracy	**05**		
Surface finish	**05**		
Report writing	**05**		
Viva	**05**		
Total Marks	**20**		

Spur Gear Cutting using Milling Machine

Experiment
08





All Dimensions are in mm

Indexing Plate
P1 = 15, 16, 17, 18, 19, 20
P2 = 21, 23, 27, 29, 31, 33
P3 = 37, 39, 41, 43, 47, 49

Calculations:

1. Outer diameter = m [Z + 2]

2. Pitch Circle diameter = mZ

3. Circular Pitch = πm

4. Tooth Thickness = πm / 2

5. Addendum = m

6. Deddendum = 1.25 m

7. Working Depth = 2.25 m

8. Indexing = $\frac{40}{Z}$

Inference:

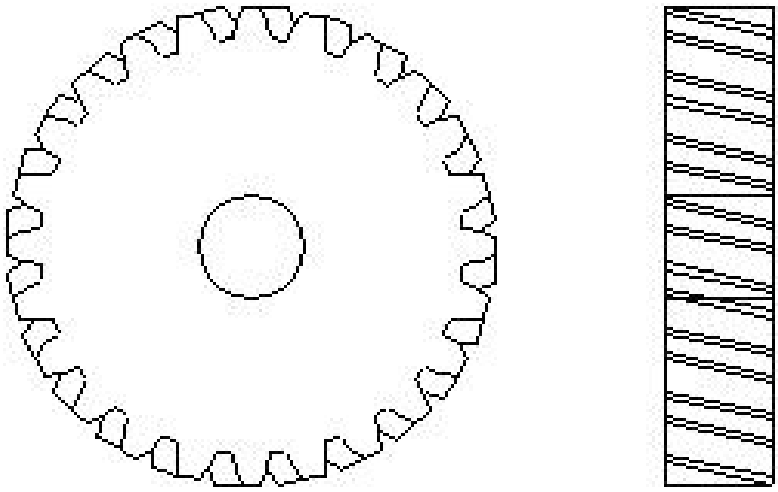
Scheme of Evaluation:

Particulars	Marks	Marks obtained	Staff Signature
Dimensional Accuracy	**05**		
Surface finish	**05**		
Report writing	**05**		
Viva	**05**		
Total Marks	**20**		

Experiment

09

Helical Gear Cutting using Milling Machine



Calculations:

1. Outer diameter = m [Z + 2]

2. Pitch Circle diameter = mZ

3. Circular Pitch = πm

4. Working Depth = 2.25 m

5. Indexing = $\frac{40}{Z}$

6. Tan θ = $\frac{\pi d}{l}$ where L = 450

7. Change gear ratio = $\frac{\text{Lead of the machine (l)}}{\text{Lead to be cut (L)}}$

Inference:

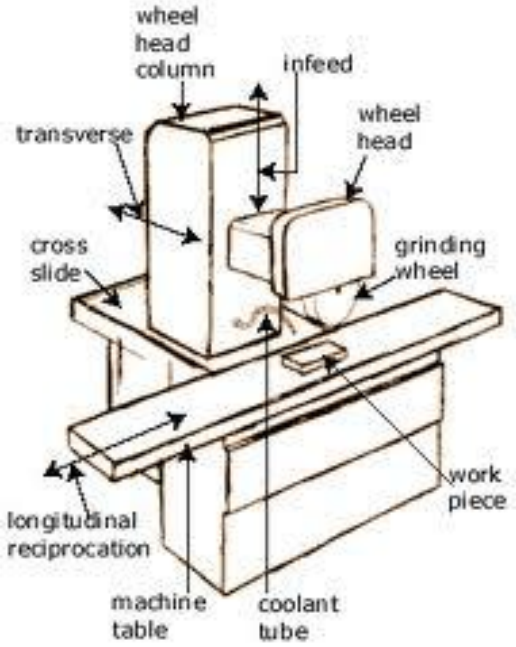
Scheme of Evaluation:

Particulars	Marks	Marks obtained	Staff Signature
Dimensional Accuracy	**05**		
Surface finish	**05**		
Report writing	**05**		
Viva	**05**		
Total Marks	**20**		

Experiment

10

Demonstration of Surface Grinding Machine



Process: Surface grinding is the most common of the grinding operations. It is a finishing process that uses a rotating abrasive wheel to smooth the flat surface of metallic or nonmetallic materials to give them a more refined look or to attain a desired surface for a functional purpose.

The surface grinder is composed of an abrasive wheel, a work holding device known as a chuck, and a reciprocating or rotary table. The chuck holds the material in place while it is being worked on. It can do this one of two ways: ferromagnetic pieces are held in place by a magnetic chuck, while non-ferromagnetic and nonmetallic pieces are held in place by vacuum or mechanical means. A machine vise (made from ferromagnetic steel or cast iron) placed on the magnetic chuck can be used to hold non-ferromagnetic work pieces if only a magnetic chuck is available.

Factors to consider in surface grinding are the material of the grinding wheel and the material of the piece being worked on.

Typical work piece materials include cast iron and mild steel. These two materials don't tend to clog the grinding wheel while being processed. Other materials are aluminum, stainless steel, brass and some plastics. When grinding at high temperatures, the material tends to become weakened and is more inclined to corrode. This can also result in a loss of magnetism in materials where this is applicable.

The grinding wheel is not limited to a cylindrical shape and can have a myriad of options that are useful in transferring different geometries to the object being worked on. Straight wheels can be dressed by the operator to produce custom geometries. When surface grinding an

object, one must keep in mind that the shape of the wheel will be transferred to the material of the object like a mirror image.

Spark out is a term used when precision values are sought and literally means "until the sparks are out (no more)". It involves passing the work piece under the wheel, without resetting the depth of cut, more than once and generally multiple times. This ensures that any inconsistencies in the machine or work piece are eliminated.

Equipment:
A surface grinder is a machine tool used to provide precision ground surfaces, either to a critical size or for the surface finish.

The typical precision of a surface grinder depends on the type and usage, however +/- 0.002 mm (+/- 0.0001") should be achievable on most surface grinders.

The machine consists of a table that traverses both longitudinally and across the face of the wheel. The longitudinal feed is usually powered by hydraulics, as may the cross feed, however any mixture of hand, electrical or hydraulic may be used depending on the ultimate usage of the machine (i.e.: production, workshop, cost). The grinding wheel rotates in the spindle head and is also adjustable for height, by any of the methods described previously. Modern surface grinders are semi-automated, depth of cut and spark-out may be preset as to the number of passes and once setup the machining process requires very little operator intervention.

Depending on the work piece material, the work is generally held by the use of a magnetic chuck. This may be either an electromagnetic chuck, or a manually operated, permanent magnet type chuck; both types are shown in the first image.

The machine has provision for the application of coolant as well as the extraction of metal dust (metal and grinding particles).

Types of surface grinders:

Horizontal-spindle (peripheral) surface grinders: The periphery (flat edge) of the wheel is in contact with the workpiece, producing the flat surface. Peripheral grinding is used in high-precision work on simple flat surfaces; tapers or angled surfaces; slots; flat surfaces next to shoulders; recessed surfaces; and profiles.

Vertical-spindle (wheel-face) grinders: The face of a wheel (cup, cylinder, disc, or segmental wheel) is used on the flat surface. Wheel-face grinding is often used for fast material removal, but some machines can accomplish high-precision work. The workpiece is held on a reciprocating table, which can be varied according to the task, or a rotary-table machine, with continuous or indexed rotation. Indexing allows loading or unloading one station while grinding operations are being performed on another.

Disc grinders and double-disc grinders: Disc grinding is similar to surface grinding, but with a larger contact area between disc and work piece. Disc grinders are available in both vertical and horizontal spindle types. Double disc grinders work both sides of a workpiece simultaneously. Disc grinders are capable of achieving especially fine tolerances.

Lubricants:

Lubricants are sometimes used to cool the workpiece and wheel, lubricate the interface and remove scarf (chips). It must be applied directly to the cutting area to ensure that the fluid is not carried away by the grinding wheel. Common lubricants include water-soluble chemical fluids, water soluble oils, synthetic oils, and petroleum-based oils. The type of lubrication used depends on the work piece material and is outlined in the table below.

Types of lubricants used for grinding based on work piece material

Work piece material	Lubricant
Aluminium	Light duty oil
Brass	Light duty oil
Cast iron	Heavy duty emulsifiable oil, light duty chemical and synthetic oil
Mild steel	Heavy duty water-soluble oil
Stainless steel	Heavy duty emulsifiable oil, heavy duty chemical and synthetic oil
Plastics	Water-soluble oil, dry, heavy duty emulsifiable oil, light duty chemical and synthetic oil

Effects on work material properties:

The high temperatures encountered at the ground surface create residual stresses and a thin martensitic layer may form on the part surface; this decreases the fatigue strength. In ferromagnetic materials, if the temperature of the surface is raised beyond the Curie temperature then it may lose some magnetic properties. Finally, the surface may be more susceptible to corrosion.

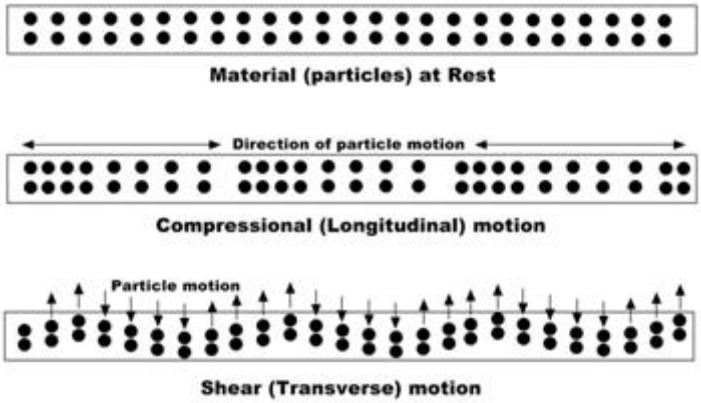
Demonstration of Non Destructive Testing Equipment's (Ultrasonic & Magnetic Particle Inspection)

Nondestructive testing (NDT) is the process of inspecting, testing, or evaluating materials, components or assemblies for discontinuities, or differences in characteristics without destroying the serviceability of the part or system. In other words, when the inspection or test is completed the part can still be used. Today modern nondestructive tests are used in manufacturing, fabrication and in-service inspections to ensure product integrity and reliability, to control manufacturing processes, lower production costs and to maintain a uniform quality level. During construction, NDT is used to ensure the quality of materials and joining processes during the fabrication and erection phases, and in-service NDT inspections are used to ensure that the products in use continue to have the integrity necessary to ensure their usefulness and the safety of the public.
It should be noted that while the medical field uses many of the same processes, the term "nondestructive testing" is generally not used to describe medical applications.

Ultrasonic Testing:
Ultrasonic testing uses the same principle as is used in naval SONAR and fish finders. Ultra-high frequency sound is introduced into the part being inspected and if the sound hits a material with a different acoustic impedance (density and acoustic velocity), some of the sound will reflect back to the sending unit and can be presented on a visual display. By knowing the speed of the sound through the part (the acoustic velocity) and the time required for the sound to return to the sending unit, the distance to the reflector (the indication with the different acoustic impedance) can be determined. The most common sound frequencies used in UT are between 1.0 and 10.0 MHz, which are too high to be heard and do not travel through air. The lower frequencies have greater penetrating power but less sensitivity (the ability to "see" small indications), while the higher frequencies don't penetrate as deeply but can detect smaller indications.

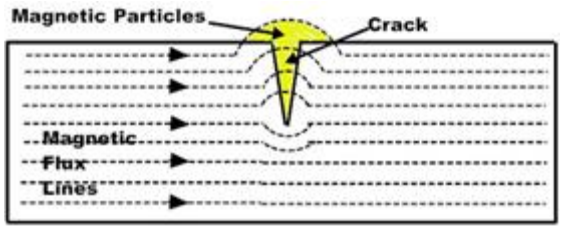
The two most commonly used types of sound waves used in industrial inspections are the compression (longitudinal) wave and the shear (transverse) wave, as shown in Figure 10. Compression waves cause the atoms in a part to vibrate back and forth parallel to the sound direction and shear waves cause the atoms to vibrate perpendicularly (from side to side) to the direction of the sound. Shear waves travel at approximately half the speed of longitudinal waves.

Sound is introduced into the part using an ultrasonic transducer ("probe") that converts electrical impulses from the UT machine into sound waves, then converts returning sound back into electric impulses that can be displayed as a visual representation on a digital or LCD screen (on older machines, a CRT screen). If the machine is properly calibrated, the operator can determine the distance from the transducer to the reflector, and in many cases, an experienced operator can determine the type of discontinuity (like slag, porosity or cracks in a weld) that caused the reflector. Because ultrasound will not travel through air (the atoms in air molecules are too far apart to transmit ultrasound), a liquid or gel called "coolant" is used between the face of the transducer and the surface of the part to allow the sound to be transmitted into the part.



Magnetic Particle Testing:

Magnetic Particle Testing uses one or more magnetic fields to locate surface and near-surface discontinuities in ferromagnetic materials. The magnetic field can be applied with a permanent magnet or an electromagnet. When using an electromagnet, the field is present only when the current is being applied. When the magnetic field encounters a discontinuity transverse to the direction of the magnetic field, the flux lines produce a magnetic flux leakage field of their own as shown in Figure 1. Because magnetic flux lines don't travel well in air, when very fine colored ferromagnetic particles ("magnetic particles") are applied to the surface of the part the particles will be drawn into the discontinuity, reducing the air gap and producing a visible indication on the surface of the part. The magnetic particles may be a dry powder or suspended in a liquid solution, and they may be colored with a visible dye or a fluorescent dye that fluoresces under an ultraviolet ("black") light.



Experiment

11

Project based learning:

Topic:

Scheme of Evaluation:

Particulars	Marks	Marks obtained	Staff Signature
Conceptual Understanding of new technologies	**10**		
Inference about the process	**10**		
Total Marks	**20**		

Sample Viva Question with answers

1. What is core sand?

   Sand that prepared by mixing a special binder such as core oil with sand, that develop high strength.

2. What are the properties of good moulding sand?
   a. Permeability
   b. Cohesiveness or Strength
   c. Adhesiveness
   d. Plasticity
   e. Refractoriness
   f. Binding
   g. Chemical resistivity
   h. Flowability.

3. What is molding sand?

   Its prepared by using a mixture of base sand, binder, additives and with or without water. The mixture of sand is referred to as moulding sand.

4. What is a binder?

   It is a material, which imparts the necessary binding action to the sand. A binder holds the sand grains together and induces strength and other properties to the moulds.

5. What is green sand?

Containing 5-7% water and 6-10% clay it's the cheapest of sand mix.

6. What is permeability?

   The ability of sand to allow the gasses to escape from the mould

7. What is clay?

Its most common binder system used in foundries, it's a product of siliceous rocks, containing hydro silicates of alumina (Al2o3).

8. What is the importance of GFN? Explain how GFN affects the properties of sand and casting.

Grains fineness number Fine grain sand result in good surface, on the casting but gases cannot escape out of the mould made from it. Coarse grain sand allows gases to escape out easily but the casting surface will be very rough. Hence grain size should select appropriately.

9. What are the different types of patterns used in foundry?
   a. Solid pattern
   b. Split pattern
   c. Cope and Drag pattern
   d. Loose piece pattern
   e. Gated Pattern
   f. Match plate pattern
   g. Follow board pattern
   h. Skeleton pattern Sweep pattern.

10. What are the different methods of preparing moulds?

a. Open mould Pit mould
b. Loma mould
c. Sweep mould
d. Core sand mould
e. Cement bonded mould
f. Plaster mould
g. CO_2 Sand mould
h. Shell mould.

11. What is a dry sand mould? Why binder is used.
It refers to sand mould in the dried condition when there is no moisture in the mould, dextrin is used up to 2% with clay as Binder.

12. What are the different types of furnaces used for melting?
a. Crucible Furnace
b. Coke Fired Furnace
c. Gas or oil Fired Furnace
d. Resistance Furnace
e. Electric Arc Furnace
f. Induction Furnace
g. Cupola Furnace.

13. What are the different tools used for making moulds in the foundry?
a. Round Rammer
b. English Trowels
c. English cleaner
d. Spoon tool
e. Smoothers -Bellows.

14. What is pattern allowance? Why it is required.
Pattern has the same shape as that of the casting but the dimensions will be generally than that of the casting. This extra dimensions or deviations from the required value, given on the pattern are referred to as pattern allowance.

15. Name different allowances given for the pattern.
a. Shrinkage allowances
b. Draft allowances
c. Machining allowances
d. Scale & grinding allowances
e. Distortion allowances

16. What are the different patterns materials used in foundry?
a. Wood
b. Metal
c. Plastic
d. Wax
e. Plaster of Paris.

17. What is cope and Drag?

The upper part of mould is cope and lower part of mould is drag.

18. What is foundry?
Shaping the metal components in their molten stage is known as Foundry.

19. Name different additives used in foundry practice.
    a. Cereal husks
    b. Sea Coal
    c. pitch
    d. wood flour
    e. Silica flour
    f. Iron oxide
    g. Dextrin
    h. Molasses.

20. What is casting?
Casting is a manufacturing process by which a liquid material is usually poured into a mold, which contains a hollow cavity of the desired shape, and then allowed to solidify. The solidified part is also known as a casting, which is ejected or broken out of the mold to complete the process.

21. Name the different casting processes.
    a. Sand Casting
    b. Permanent Mold Casting
    c. Low Pressure Casting
    d. Die Casting (high pressure)
    e. Direct Squeeze Casting
    f. Continuous Casting Investment Casting
    g. Shell mould casting
    h. Vacuum Casting
    i. Investment (top) and Conventional Castings
    j. Lost Foam Casting
    k. Centrifugal Casting

22. What is a Core?
Core is sand blocks they are used to make hollow portion in a casting.

23. How a core is made?
Core is made separately in a core box made of wood or metal.

24. What is Sprue?
It is taper shaped vertical component in the system. It will help transfer of molten metal from the pouring cup into the runner.

25. What is Runner and Riser?
It is a long horizontal channel which carries molten metal and distributes it to the ingates or gates. It will ensure proper supply of molten metal to the cavity so that proper filling of the cavity takes place. Riser is to supply the molten metal to the solidifying casting and eliminate shrinkage cavity and should permit easy escape of air and gases.

26. Name some of the Common casting defects.
    a. Porosity/Holes
    b. Shrinkage cavity
    c. Inclusions
    d. Hot Tears -Scab, Rat tails
    e. Sand burn-on/sand fusion
    f. Cold shut
    g. Core shift/Mould shift
    h. Fins.

27. What is gating?
    Molten metal is conveyed into the mould cavity and the mould gets completely filled up. After solidification, the solid metal represents casting, For conveying the molten metal into the mould cavity components referred to as Gating system.

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